11 DATA INTERPRETATION

11.1. Data Interpretation

- It deals with careful reading, understanding, organizing and interpreting the data provided so as to derive meaningful conclusions.
- Mostly used tools for interpretation of a data are
  - Ratio
  - Percentage
  - Rate
  - Average

11.2. Types of Data Interpretation:

The numerical data pertaining to any event can be presented by any one or more of the following methods.

1. Tables
2. Line Graphs
3. Bar Graphs or Bar Charts
4. Pie Charts or Circle Graphs

11.2.1. Tables

It is the systematic presentation of data in tabular form to understand the given information and to make clear the problem in a certain field of study. It has six elements namely:

- **Title:** It is the heading of the table.
- **Stule:** It is the section of the table containing row headings
- **Column Captions:** It is the heading of each column
- **Body:** It consists the numerical figures
- **Footnotes:** It is for further explanation of the table
- **Source:** It is the authority of the data

Example: Study the following table and answer the questions given below it.

Annual Income of Five Schools

Figures in '00 rupees

Sources of Income	School A	School B	School C	School D	School E
Tuition Fee	120	60	210	90	120
Term Fee	24	12	45	24	30
Donations	54	21	60	51	60
Funds	60	54	120	42	55
Miscellaneous	12	3	15	3	15
Total	270	150	450	210	280

Example: The income by way of donation to school D is what per-cent of its miscellaneous?

Solution: Required percentage = $\frac{5100}{300} = 27\%$

11.2.2. Line Graph

A line graph indicates the variation of a quantity w.r.t two parameters calibrated on X and Y-axis respectively.

Note:

1. Any part of the line graph parallel to X-axis represents no change in the value of Y parameter w.r.t the value of X parameter.
2. The steepest or maximum part of the line graph indicates maximum percentage change of the value during the two consecutive period in which the related part lies.
3. If the steepest part is a rise slope, then it is the highest percentage growth.
4. If the steepest part is a decline slope, it will represent a maximum percentage fall of the value calibrated in the other axis.

11.2.3. Bar Graph

Bar graphs are diagrammatic representation of a discrete data.

Types of Bar Graphs:

- **Simple Bar Graphs:** A simple bar graph relates to only one variable. The values of the variables may relate to different years or different terms.
- **Sub-divided Bar Graph:** It is used to represent various parts of sub-classes of total magnitude of the given variable.
- **Multiple Bar Graphs:** In this type, two or more bars are constructed adjoining each other, to represent either different components of a total or to show multiple variables.

11.2.4. Pie Chart

In this method of representation, the total quantity is distributed over a total angle of 360° which is one complete circle or pie.

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